

STOCKHOLM ARLANDA, Sweden

WMO: 024600

Lat: 59.627N

Lon: 17.954E

Elev: 137

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	8.3	-0.8	5.3	3.9	4.2	6.8	9.6	26.7	37.8	24.0	36.8	4.4	250	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	81.3	64.2	78.1	63.1	74.9	61.5	67.1	75.8	65.2	73.6	63.5	71.5	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	91.0	69.6	62.4	84.7	68.1	60.5	79.0	66.7	31.6	75.6	30.2	73.0	28.9	71.6	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	19.5	17.4	DB	-3.0	85.1	6.7	3.7	-7.8	87.8	-11.7	89.9	-15.5	92.0	-20.4	94.7	
			WB	-3.3	69.3	6.7	2.5	-8.0	71.1	-11.9	72.6	-15.7	74.0	-20.5	75.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	27.2	27.4	32.7	41.8	50.7	58.4	64.0	61.9	53.8	44.2	36.2	30.3
	DBStd	14.71	9.00	8.92	7.50	6.51	6.83	5.42	5.24	5.20	5.77	6.47	7.06	8.96
	HDD50	3465	705	634	537	257	76	4	0	0	29	199	414	610
	HDD65	7701	1170	1054	1001	696	445	207	84	125	336	644	864	1075
	CDD50	1328	0	0	0	11	97	255	433	368	144	20	0	0
	CDD65	90	0	0	0	0	1	9	52	27	1	0	0	0
	CDH74	844	0	0	0	0	25	118	488	209	4	0	0	0
	CDH80	146	0	0	0	0	0	11	108	27	0	0	0	0
	WSAvg	8.5	8.8	8.8	8.7	8.8	8.7	8.5	7.9	7.6	8.1	8.2	8.6	9.0
Precipitation	PrecAvg	22.0	1.5	1.2	1.1	1.2	1.5	2.2	2.6	2.6	2.0	2.1	2.0	1.8
	PrecMax	25.9	3.1	2.6	2.0	3.7	2.5	3.8	5.5	5.9	5.3	4.0	4.7	4.3
	PrecMin	17.3	0.3	0.1	0.1	0.2	0.6	0.7	0.4	0.2	0.2	1.0	0.6	0.4
	PrecStd	2.0	0.8	0.5	0.4	0.9	0.6	1.0	1.4	1.5	1.3	0.8	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	46.4	55.7	68.3	77.1	81.8	87.0	82.8	73.6	61.2	53.1	50.0
		MCWB	42.7	42.8	45.2	52.9	59.4	62.5	66.3	64.5	60.4	55.8	50.4	47.2
	2%	DB	42.5	43.0	50.8	62.7	72.9	77.3	82.8	79.2	69.6	58.0	50.2	46.2
		MCWB	40.7	39.9	42.6	49.8	57.1	61.0	64.8	64.6	59.6	54.1	48.1	44.4
	5%	DB	40.4	40.8	46.6	58.7	68.4	73.7	79.3	75.6	66.3	55.8	48.2	43.2
		MCWB	38.8	38.2	40.6	48.1	54.9	59.1	63.9	63.4	58.1	52.5	46.6	41.8
	10%	DB	38.3	38.8	43.0	54.0	64.2	70.1	75.6	72.5	63.2	53.8	46.2	41.1
		MCWB	36.9	36.7	38.4	44.8	52.1	57.8	62.4	61.7	56.7	50.9	44.8	39.8
	0.4%	WB	43.3	43.5	46.6	54.5	62.5	65.7	70.2	70.1	62.9	57.4	51.3	47.4
		MCDB	44.1	45.3	53.8	65.0	74.0	76.5	79.7	77.6	69.1	60.0	52.5	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	40.3	43.4	51.0	59.0	63.2	67.6	67.0	60.8	54.9	48.5	44.2
		MCDB	41.4	42.2	49.0	60.6	68.1	72.9	77.2	74.2	66.9	57.4	49.6	45.2
	5%	WB	38.6	38.4	41.1	48.6	56.1	60.9	65.8	65.1	59.3	53.4	46.5	41.9
		MCDB	39.7	40.1	45.3	57.1	65.9	70.0	74.9	71.9	64.8	55.8	47.8	42.8
	10%	WB	36.8	36.5	39.1	46.1	53.6	59.1	64.0	63.4	57.6	51.5	44.7	39.6
		MCDB	37.9	38.1	42.3	52.6	62.9	67.9	73.0	69.9	62.2	53.6	45.7	40.6
Mean Daily Temperature Range	5% DB	MDBR	8.6	10.5	13.2	17.3	18.9	17.5	17.7	16.9	15.0	11.3	8.1	8.1
		MCDBR	8.3	10.0	17.4	24.5	26.1	24.3	23.8	21.5	18.5	12.5	9.8	8.4
		MCWBR	7.9	8.0	11.3	13.9	13.4	11.4	9.5	9.3	10.1	9.1	8.9	8.0
	5% WB	MCDBR	8.0	9.1	15.6	22.3	22.6	20.5	19.8	17.8	15.9	11.3	9.1	7.8
		MCWBR	7.9	7.8	11.0	13.1	12.5	10.7	9.2	8.9	9.3	8.9	8.5	7.6
Clear-Sky Solar Irradiance	taub		0.278	0.318	0.337	0.373	0.370	0.361	0.388	0.379	0.345	0.333	0.304	0.262
	taud		2.207	2.279	2.353	2.324	2.394	2.449	2.395	2.416	2.498	2.419	2.251	2.129
	Ebn at Noon		162	217	252	262	272	277	265	259	250	212	149	119
	Edh at Noon		14	22	27	34	34	32	33	30	24	19	13	10
All-Sky Solar Radiation	RadAvg		92	280	758	1259	1657	1848	1708	1347	869	382	120	61
	RadStd		13	45	88	119	145	112	164	123	95	62	27	7

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+3.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (51 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	-214	-265	N/A	N/A	

Nomenclature: See separate page